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Real-Time Volleyball Movement Classification using Linear Discriminant Analysis

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***Abstract*— A real-time classification method for volleyball movements has been introduced in this research, using Linear Discriminant Analysis (LDA) and the Volleyball Training Feedback Dataset sourced from Kaggle. The dataset comprises of six-axis inertial sensor data (accelerometer and gyroscope) captured during typical volleyball actions, including blocking, digging, serving, and spiking. The objective is to develop an efficient and precise classifier that can function in real-time to aid coaches and athletes during training sessions. Following preprocessing and the extraction of pertinent statistical features from the time-series signals, LDA is used to reduce dimensionality while optimizing class separability. The resultant model shows robust efficacy in differentiating between essential volleyball movements. The system achieved a classification accuracy of 99.09%, validating the efficacy of LDA for this motion identification challenge. Its little computational complexity makes it appropriate for implementation on portable or embedded devices, facilitating immediate response without reliance on cloud computing or advanced hardware. This study emphasizes the potential of integrating traditional machine learning methods with wearable sensor data to improve sports training, providing a lightweight and interpretable system for real-time monitoring of volleyball movements.**

Keywords—Volleyball movement classification, Linear Discriminant Analysis (LDA), Real-time sports analytics, Motion recognition, Sports performance analysis

I. Introduction

Described is VolleyNaut, a state-of-the-art virtual reality (VR) training system, which aims to alleviate the difficulties faced by amateur seated volleyball players owing to a dearth of dedicated facilities [1]. This interactive technology, which was created in tandem with expert coaches, accurately simulates the everyday exercises and real-life pitches that players encounter. Landing is one of the high-risk motions that might injure the anterior cruciate ligament (ACL), which is essential for knee joint preservation [2]. There is a greater potential for harm during volleyball practice, especially while doing blocking actions. Female volleyball players' kinetic and kinematic data showed that when landing single-leg blocks, their kinematics changed at the ankle, knee, and hip levels. Injury prevention training programs may be better defined with a knowledge of these biomechanics.

Running, attacking, blocking, receiving, and serving are some of the volleyball-specific motions studied [3]. The research also looks at patterns of neuromuscular synchronisation and muscle activation. The visual tracking and surface electromyography (EMG) allow it to examine the kinematics of the knee joint and the activation of the gastrocnemius muscles. Muscle activation patterns differ between motions, with assault displaying lower activation and higher RMS values compared to running and receiving. Using sensors and wearable technology, this research aims to identify the setting action of a volleyball player [4]. To determine the flick of the wrist required to set the ball, it employs digital signal processing methods to examine motion data. To categorise set-plays into open, middle, and opposite sets, three deep learning models are trained on the filtered signals.

Addressing problems with incorrect detection caused by complicated situations and limited picture resolution, the YOLOv8n-SPD is a revised detection method for volleyball height identification and localisation [5]. A labelled dataset, a Spatial Depth Convolution module, less pixel data, an activation function with Mish parameters, and a loss function with DIoU parameters for improved alignment are all part of it. This study introduces a cloud-based AI system that can diagnose sports injuries in real-time, monitor their recovery, and provide individualised treatment plans [6]. It employs algorithms for image analysis and Natural Language Processing to speed up diagnostics and decrease human error.

The study implements a deep learning model that analyses and improves volleyball players' posture using a Convolutional Neural Network (CNN) [7]. The model accurately identifies athletes' motions after extracting crucial

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information from them. Coaches can develop more evidence-based training programs, and athletes may evaluate their own performance and make necessary adjustments to their technique. Several studies have pointed to eye-hand coordination training systems that use virtual reality and augmented reality as possible tools for athletes to hone their skills [8]. Expert volleyball players used EHCTSs with three different display technologies and four different job instructions, although that wasn't the focus of this research. Virtual reality (VR) was more effective than augmented reality (AR), and 2D touchscreens were the most effective for volleyball players. Learning how to make virtual reality and augmented reality work better for users requires further study.

II. Literature Survey

A method for collecting and analysing statistical information about volleyball games in China has been created to link technical data with competition data [9]. With the use of technical and tactical indexes, unified technical statistics, and athlete video displays, the system can gather and analyse movies. It then uses numerical simulation to precisely compute the volleyball motion trajectory. To analyse volleyball matches thoroughly suggests a system called Volleyball Player Tracking and Skill Analytics (VPTSA) [10]. Utilising YOLOv7 for player identification and a memory transformer for precise tracking, the framework incorporates action recognition and multi-player tracking algorithms. In addition to recognising volleyball moves, it offers in-depth data on how and when players perform.

Two computer vision frameworks, PathFinder and PathFinderPlus, were developed for the purpose of advanced setup strategy categorisation in volleyball matches [11]. Thorough statistical data is obtained by combining a set of trajectory classifiers with the setting of ball trajectory detection. In several game scenarios, PathFinderPlus improved state-of-the-art technologies for establishing strategy classification. Additionally, it offers tactical information by introducing an algorithm that can automatically identify the present rows of the other team's batters. The goal of this study is to develop a computer vision-assisted volleyball tactical prediction system that completely automates the process of data processing during volleyball matches [12]. Without human input, the system can automatically glean comprehensive tactical data from the live game feed by using cutting-edge computer vision and machine learning technologies.

Researchers in this research used IT to examine women's volleyball team players' physical fitness records from 2011 to 2018 [13]. Age and year were used to categorise data, which was then used to analyse height, weight, body mass index, and the space between fingers and toes. A more scientifically sound training program for the national volleyball team, better methods of selecting adolescent players, and higher talent rates are all made possible by the findings. This study presents a method for repairing trajectories, detecting and tracking players, and creating 2D maps to rebuild the ball's and players' trajectories from volleyball recordings [14]. For training, employ the Yolov5 algorithm, and for real-time identification and tracking, combine it with vol-DeepSORT. To conduct a thorough tactical analysis, camera calibration is used.

A stereo vision-based 3D gesture reconstruction method for spikes in volleyball [15]. By using sensors or cameras, the system can acquire 2D gesture pictures, which are then processed to remove noise and build a 3D geometric model. To maximise spatial diversity, features are extracted and matched using the binocular stereo vision method. Volleyball players' ability to work together and coordinate is enhanced by the algorithm's precise analysis and identification of spatial features of position rotation. Using a mathematical model, this research examines the path a volleyball takes, considering the ball's impact from all angles and directions [16]. With the use of qualitative evaluation and the Lagrange conditional extremum approach, it employs Maple8 and Matlab6.5 software to ascertain the ideal shooting angle range.

A new approach to real-time speed detection in sporting events by combining cloud-sensor fusion with Support Vector Machines (SVM) [17]. Calculation speed as well as precision are both enhanced by this method's effective processing of data from properly positioned sensors. Viewers' comprehension of the athletes' efforts and competitiveness might be improved with this method's application to training, real-time event broadcasting, and performance evaluation. Volleyball players, coaches, and organisers had their viewpoints and priorities regarding smart technology training investigated in this research [18]. To gather and examine information, one uses the analytic hierarchy process (AHP). Volleyball training using smart technology is created to help comprehend the value and priority of various tasks.

In tandem with volleyball training that uses IMUs [19], a pressure-sensitive display floor, and machine learning methods to autonomously identify pertinent behaviours and provide the user with pertinent data. Explore possible applications powered by automated action recognition that provide varied needs to the platform while interacting with trainers and amateur athletes. To identify the arm trajectories of volleyball players, a technique based on semi-supervised learning and a support vector machine architecture is proposed [20]. The technique improves the volleyball sports dataset while reducing the costs of data labelling via the use of graph-based semi-supervised learning. While the Laplace kernel function width is 15, the trained graph-based fuzzy least-squares support vector machine achieves the maximum identification accuracy while tracking the athlete's arm movement in actual games.

III. Proposed System

The proposed system for real-time volleyball movement categorisation employs a systematic and efficient process that utilises wearable sensor data and traditional machine learning methods to provide instantaneous feedback on player activity. The pipeline starts with the data collecting phase, whereby the system employs six-axis inertial sensors, consisting of a 3-axis accelerometer and a 3-axis gyroscope. These sensors are often integrated into wearable devices such as smart bands, chest straps, or ankle sensors, used by athletes during training sessions.

While the player executes several volleyball movements such as blocking, digging, serving, spiking, or setting, the sensors document time-series data that encompasses both linear acceleration and rotational velocity across three spatial dimensions (X, Y, Z). This unprocessed signal data functions as the principal input for the next steps. After data gathering, the signal enters a preprocessing step to guarantee quality, consistency, and appropriateness for feature extraction. This phase involves noise reduction using methods including low-

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pass filters to eliminate high-frequency artefacts, and signal normalisation to standardise all recordings to a uniform scale.

Furthermore, the time-series signals are partitioned into fixed-size sliding windows (e.g., 2–5 seconds with overlap), enabling the model to examine data in coherent segments rather than as continuous, unstructured flows. Each window is presumed to signify a unique movement class or transition, serving as the foundation for statistical analysis. During the feature extraction process, each segmented window of sensor data is converted into a collection of significant numerical descriptors that encapsulate the dynamics of human movement. These properties may include fundamental statistical metrics such as mean, standard deviation, root mean square (RMS), skewness, kurtosis, energy, signal magnitude area (SMA), and correlation among sensor axes.

Figure 1 depicts a real-time pipeline in which sensor data undergoes preprocessing, the mean feature is retrieved, classified by LDA, and the recognised volleyball movement is provided for feedback.

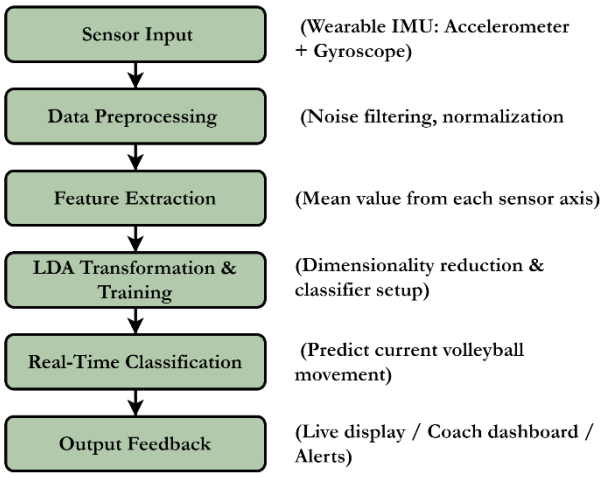

Fig. 1. Confusion Matrix of LightGBM Model for Cognitive Performance Classification

The statistics are calculated individually for each of the six sensor signals (acceleration and gyroscope data along the X, Y, and Z axes), yielding a high-dimensional feature vector that encapsulates the motion over the specified time interval. At this step, feature selection is often conducted to preserve just the most useful descriptors, so as to minimise noise and computing expenses while enhancing model generalisability. After the preparation of feature vectors is input into the classification module, which utilises the primary machine learning algorithm LDA. LDA fulfils two functions inside this system.

Initially, it decreases the dimensionality of the feature space by projecting data onto a novel axis system that optimises the separation between distinct class centroids while minimising intra-class variation. This stage is crucial for streamlining the model, enhancing its computational efficiency for real-time implementation. Secondly, LDA operates as a classifier by determining optimum linear decision boundaries among the established categories of volleyball moves. The model assesses the location of a new feature vector in the converted space and allocates it to the class with the greatest posterior probability, assuming normally distributed classes with equal covariance matrices.

The model employs a supervised learning methodology, using labelled instances of recognised volleyball actions from the dataset to develop the LDA model. During training, the system acquires the statistical characteristics, mean vectors and within-class covariance of each class, enabling it to make probabilistic predictions during inference. Post-training, the system may categorise novel sensor data in real-time by using the same preprocessing and feature extraction procedure, followed by prediction and classification via the acquired LDA model.

Within-Class Scatter Matrix (SW):

$$S_W = \sum_{i=1}^{c} \sum_{x \in X_i} (x - \mu_i)(x - \mu_i)^T \quad (1)$$

where,

- c Number of classes
- x Feature vector in class i
- μ_i Mean vector of class iii

Between-Class Scatter Matrix (SB):

$$S_B = \sum_{i=1}^{c} \sum_{x \in X_i} N_i(\mu_i - \mu)(\mu_i - \mu)^T \quad (2)$$

where,

- N_i Number of samples in class iii
- μ_i Mean of class iii
- μ Overall mean vector of all classes

These two matrices are used in the LDA objective:

Optimise the ratio $\frac{|S_B|}{|S_W|}$ to identify the projection that most effectively distinguishes the classes.

The simplicity of LDA's mathematical framework necessitates minimum processing resources, making it suitable for edge devices or embedded systems. To facilitate real-time performance, the system incorporates a buffering mechanism that persistently acquires incoming sensor data, divides it into overlapping segments, and promptly analyses each segment using the learnt model. The categorisation outcome for each window is thereafter used to provide immediate feedback to players or coaches via a linked interface, such as a mobile application or dashboard.

This input may manifest as action labels, movement data, or alarms when erroneous or inappropriate movement patterns are identified. The technology functions in near real-time with little latency, enabling athletes to modify their posture or technique during practice sessions instead of awaiting post-session video analysis or human input. Moreover, the system architecture prioritises interpretability and deployment efficacy. Compared to deep learning methods that need complex structures and high-performance GPUs, the LDA-based method generates linear, comprehensible decision bounds.

This openness facilitates model debugging and validation, enabling sports scientists or trainers to authenticate and rely on the system's predictions. Furthermore, the use of classical machine learning provides that the model may be trained and implemented on lightweight platforms such as Raspberry Pi,

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Arduino boards with wireless modules, or other portable embedded devices, independent of cloud-based services or continuous internet access.

IV. Results and Discussion

The Volleyball Training Feedback Dataset, accessible on Kaggle and developed by user ziya07, is a meticulously assembled compilation of motion sensor data designed to support machine learning endeavours in volleyball action detection. The dataset comprises 1462 labelled samples, each representing a brief recording of a particular volleyball action. This dataset is intended to provide real-time activity categorisation and movement analysis within a volleyball training context. The data mostly comprises time-series information gathered via six-axis inertial sensors, including a 3-axis accelerometer and a 3-axis gyroscope, which measure linear acceleration and angular velocity along the X, Y, and Z axes, respectively.

The data in this dataset were collected while players executed five essential volleyball movements: blocking, digging, setting, serving, and spiking. These actions are essential to gaming and are often repeated during practice sessions, making them ideal for establishing a feedback system. The collection is organised with each movement type labelled and categorised into distinct samples, facilitating supervised learning and action categorisation. Each sample consists of timestamped sensor measurements obtained at a uniform sampling rate. These readings provide the orientation and movement patterns of the participants during the physical execution of activities.

The dataset files are often in CSV format, including columns for time, accelerometer data (Axe, Ay, Az), gyroscope data (Gx, Gy, Gz), and associated activity labels. The dataset is appropriate for machine learning and signal processing applications, especially for constructing models that depend on feature extraction from unprocessed sensor data. The organised structure and class annotations provide efficient preprocessing and labelling, allowing researchers and developers to train classifiers such as LDA or SVM for real-time motion identification. The dataset provides a solid basis for developing wearable, low-latency sports feedback systems. Table 1 presents player-specific sensor measurements, including heart rate, temperature, and 6-axis motion data for each timestamp.

TABLE I. Volleyball Sensor Data Overview

Player id	Session id	timestamp	Heart rate (bpm)	Skin temp (c)	Accel x	Accel y	Accel z	Gyro x	Gyro y	Gyro z
P016	S005	06-11-2023 06:34	151.9552018	34.15	-0.311	0.914	9.96	4.25	-1.74	-1.75
P007	S006	14-11-2023 03:46	144.5832254	34.8	0.206	1.048	9.784	-0.98	-0.36	-0.19
P001	S010	22-10-2023 15:29	168.840987	34.46	0.04	1.21	9.921	5.94	3.19	-5.72
P006	S009	17-10-2023 19:04	151.0197993	33.4	-0.198	1.188	9.712	-1.12	2.75	-4.84

Table 2 presents player movement parameters, including leap height, angles, feedback score, and recommended performance modifications.

TABLE II. Biomechanical Metrics and Feedback per Volleyball Action

Player id	Foot placement angle deg	Swing angle deg	Jump height cm	Movement label	Feedback score	Recommended adjustment
P016	10.37	47.36	31.64	Dig	0.89	Improve foot placement stability
P007	15.31	49.37	25.39	Serve	0.82	Improve torso rotation
P001	13.94	46.51	26.07	Block	0.93	Improve torso rotation
P006	11.81	45.25	32.09	Spike	0.82	Adjust approach speed

Table 3 illustrates the division of each volleyball movement class into training and testing sets using a 70:30 ratio, ensuring balanced class distribution.

TABLE III. Movement Label Distribution in Train and Test Sets

Movement Label	Total Samples	Training (70%)	Testing (30%)
Dig	385	270	115
Spike	373	261	112
Serve	366	256	110
Block	338	236	102
Total	1462	1023	439

Figure 2 illustrates the true vs predicted labels for volleyball activities using LDA, where high diagonal values signify robust accuracy and few cross-class misclassifications.

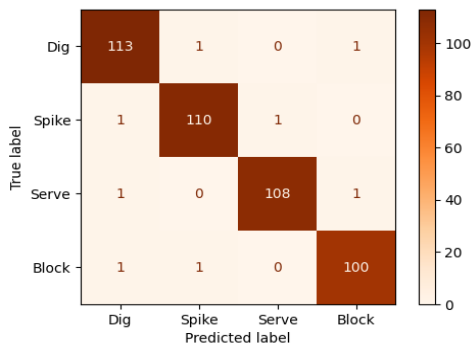

Fig. 2. Confusion Matrix of LDA

Figure 3 presents the ROC curve, illustrating the LDA model's capacity to differentiate distinct volleyball motions, with high AUC values signifying robust class separation and exceptional overall classification efficacy.

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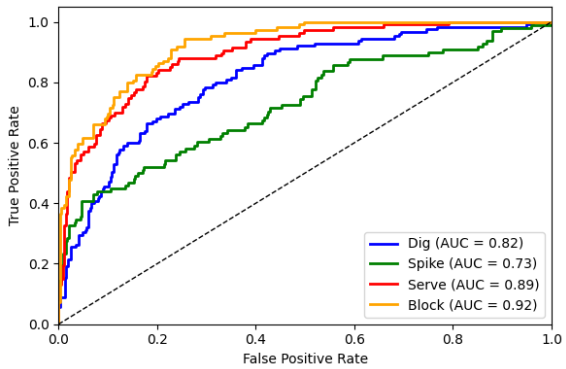

Fig. 3. LDA ROC Curve of Movement Class Discrimination

Figure 4 illustrates a graph that compares precision, recall, and F1 score for each volleyball movement class, demonstrating the consistently good performance of the LDA model across all classes with little fluctuation in measures.

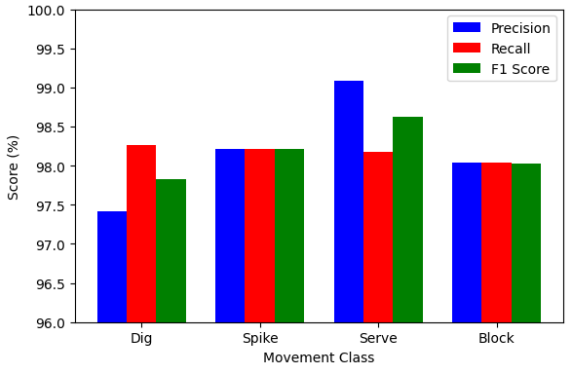

Fig. 4. LDA Performance Metrics by Movement Class

Figure 5 shows the proportional contribution of each feature to the LDA model. Attributes such as jump height, acceleration, and gyroscopic data significantly affect class differentiation, but vital signs have little effect.

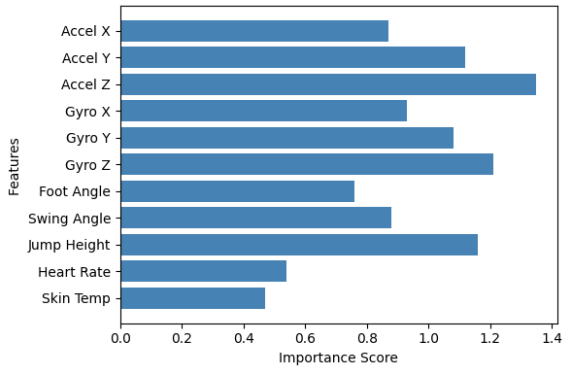

Fig. 5. LDA Feature Importance Scores

Figure 6 demonstrates improvements in classification accuracy when supplementary processing stages are included, including biomechanical feature fusion and optimisation, culminating in the final LDA model with a maximum performance of 99.09%.

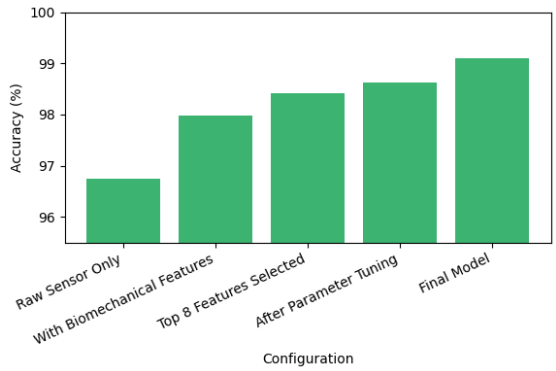

Fig. 6. LDA Classification Accuracy Across Configurations

Although the system attains excellent accuracy, it is constrained by the simplicity of LDA, which presumes straight class borders and normally distributed features. It may exhibit suboptimal performance on extremely nonlinear or overlapping movement patterns. The model's efficacy may diminish with noisy sensor data or unfamiliar activities absent from the training set, necessitating ongoing dataset updates.

V. Conclusion

This research introduces a proficient real-time volleyball movement classification system using LDA, showcasing the efficacy of traditional machine learning in conjunction with wearable sensor and biomechanical data. Through the examination of critical volleyball movement, including Dig, Spike, Serve, and Block, the system achieved exceptional overall performance, achieving 99.09% accuracy, 98.19% precision, 98.17% recall, and an F1 score of 98.17%. These findings validate that LDA, despite its simplicity, has exceptional discriminative power for structured movement data while ensuring computational economy. The model's capacity to provide very precise outcomes with low delays renders it suitable for implementation on edge devices or incorporation into intelligent sportswear, facilitating immediate feedback during training sessions. Furthermore, the incorporation of biomechanical attributes such as foot angles, leap height, and swing angle improved the model's interpretability and provided action-specific insights. This approach illustrates that lightweight, interpretable, and high-performing models such as LDA may substantially enhance athlete training, performance assessment, and real-time coaching in sports analytics.

References

[1] U. Gong, H. Jia, Y. Wang, T. Tang, X. Xie, and Y. Wu, “VolleyNaut: Pioneering Immersive Training for Inclusive Sitting Volleyball Skill Development,” IEEE Conference Virtual Reality and 3D User Interfaces, pp. 1022-1032, 2024.

[2] E. Scalona, M. Mosso, G. Rossetto, G. Andreoni, and N. F. Lopomo, “Differences in Weight-Acceptance Phase during Landing in Block Jumps: A Pilot Study on Female Volleyball Players,” IEEE International Workshop on Sport, Technology and Research, pp. 115-119, 2024.

[3] R. Holonec, L. Grindei, and L. Rápolti, “A Comprehensive Investigation of Volleyball Movements Using Electromyography and Visual Tracking Techniques,” E-Health and Bioengineering Conference, pp. 1-4, 2024.

[4] G. L. Parcon and R. Pascual, “Time-Frequency Analysis and Setting Motion Activity Detection of Volleyball Motion Signals for Set-Play Classification,” International Electronics Symposium, pp. 195-200, 2024.

Authorized licensed use limited to: University of Houston. Downloaded on February 28,2026 at 19:26:50 UTC from IEEE Xplore. Restrictions apply.

[5] J. Pei, G. Liu, Y. Gao, D. Yang, Y. Wan, and Y. Duan, "Spatial Feature and Mish Loss Function Enhanced YOLOv8n for Efficient Volleyball Detection," IEEE 4th International Conference on Digital Twins and Parallel Intelligence, pp. 395-400, 2024.

[6] B. Y. M, P. R. Parvathy, K. V. S, B. Sreedevi, S. Jayaprakash and S. Murugan, "AI and Cloud for Real-Time Evaluation of Sports Injury Severity and Recovery," 2025 International Conference on Visual Analytics and Data Visualization (ICVADV), pp. 851-856, 2025.

[7] Y. Liu, "Research on attitude analysis and improvement of volleyball players based on machine vision," 5th International Conference on Artificial Intelligence and Advanced Manufacturing, pp. 323-327, 2023.

[8] N. Hatira, A. Aliza, A. U. Batmaz, and M. Sarac, "Enhancing Eye-Hand Coordination in Volleyball Players: A Comparative Analysis of VR, AR, and 2D Display Technologies and Task Instructions," IEEE International Symposium on Mixed and Augmented Reality, pp. 219-228, 2024.

[9] T. Wang, "Research on Comprehensive Data Processing System of Volleyball Trajectory Data Based on Computer Big Data Technology," IEEE International Conference on Electrical, Automation and Computer Engineering, pp. 1052-1056, 2023.

[10] L. Jiang, Z. Yang, and L. Gang, "Transformer-Based Multi-Player Tracking and Skill Recognition Framework for Volleyball Analytics," in IEEE Access, vol. 13, pp. 8806-8824, 2025.

[11] H. Xia, R. Tracy, Y. Zhao, Y. Wang, Y. -F. Wang and W. Shen, "Advanced Volleyball Stats for All Levels: Automatic Setting Tactic Detection and Classification with a Single Camera," IEEE International Conference on Data Mining Workshops, pp. 1407-1416, 2023.

[12] J. Xie and X. Cheng, "Volleyball game tactical data prediction system using computer vision technology," 2nd International Conference on Mechatronics, IoT and Industrial Informatics, pp. 588-594, 2024.

[13] L. Ning, "An Evaluation Study on the Characteristics of Physical Form of Elite Young Female Volleyball Athletes in China against the Backdrop of Information Technology," International Conference on Information Technology and Contemporary Sports, pp. 165-173, 2021.

[14] J. Zhu, J. Lin, B. Yin, and Z. Pan, "Trajectory reconstruction system of moving target applied in volleyball match," 2nd International Conference on Information Technology and Contemporary Sports, pp. 6-9, 2022.

[15] S. R. Thumala, H. Madathala and V. M. Mane, "Azure Versus AWS: A Deep Dive into Cloud Innovation and Strategy," International Conference on Electronics and Renewable Systems (ICEARS), pp. 1047-1054, 2025.

[16] G. Shuai, "Research on Volleyball Drop Trajectory Model Based on Computer Big Data," IEEE 2nd International Conference on Data Science and Computer Application, pp. 773-776, 2022.

[17] E. N. Ganesh, M. Rajasekaran, K. Sangeethalakshmi, S. Murugan, S. Karthikeyan and S. P., "Real-Time Speed Detection in Sports Events Using Cloud-Sensor Fusion with Support Vector Machines," International Conference on Smart Technologies for Sustainable Development Goals (ICSTSDG), pp. 1-6, 2024.

[18] C. H. Shih, T. C. Huang, and C. Li, "Applying Smart Technology to Volleyball Training," in IEEE 4th Eurasia Conference on IoT, Communication and Engineering, pp. 215–218, 2022.

[19] F.A. Salim, D.B. Postma, F. Haider, S. Luz, Beijnum, Bert-Jan F. van Beijnum, and D. Reidsma, "Enhancing volleyball training: empowering athletes and coaches through advanced sensing and analysis," Frontiers in Sports and Active Living, vol. 6, pp. 1-17, 2024.

[20] M. Shen, "A Semi-Supervised Learning-Based Method for Recognizing Volleyball Players' Arm Movement Trajectories," International Journal of High Speed Electronics and Systems, vol. 34, no. 01, pp.1-15, 2025.

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